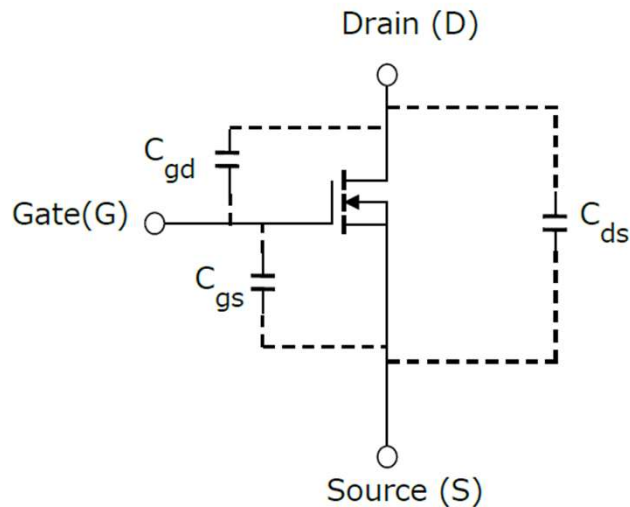




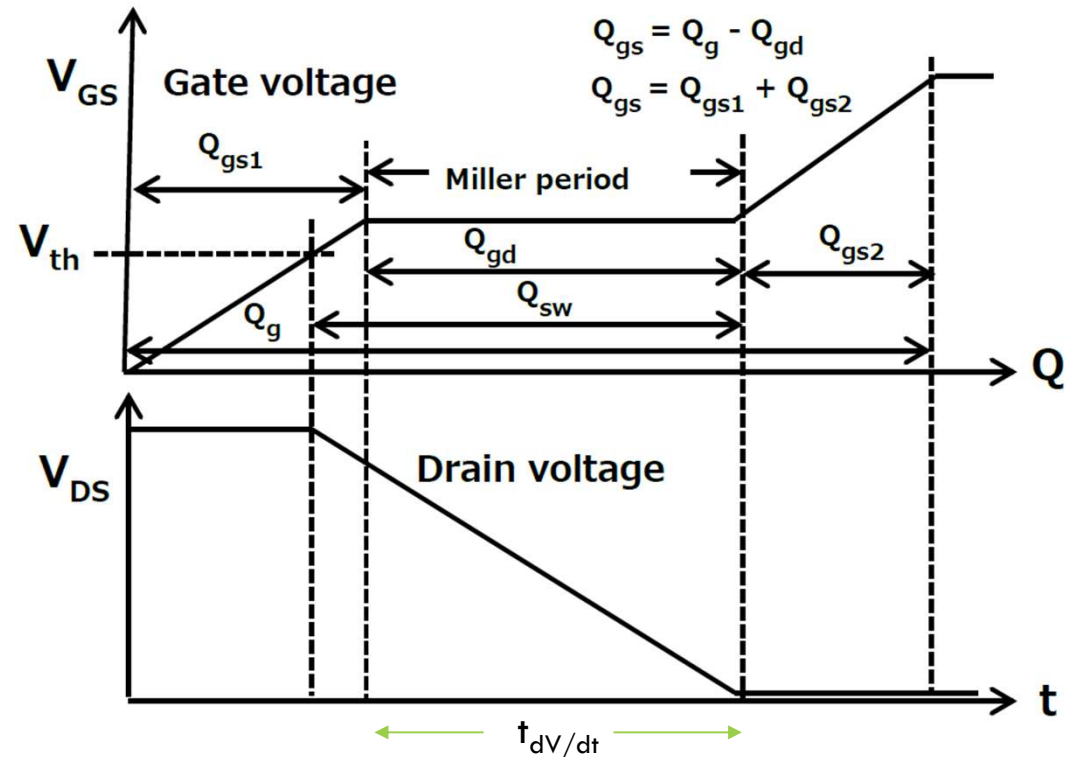
Gate Driver Fundamentals

GATE DRIVE



Input capacitance $C_{iss} = C_{gd} + C_{gs}$
 Output capacitance $C_{oss} = C_{ds} + C_{gd}$
 Reverse Transfer capacitance $C_{rss} = C_{gd}$

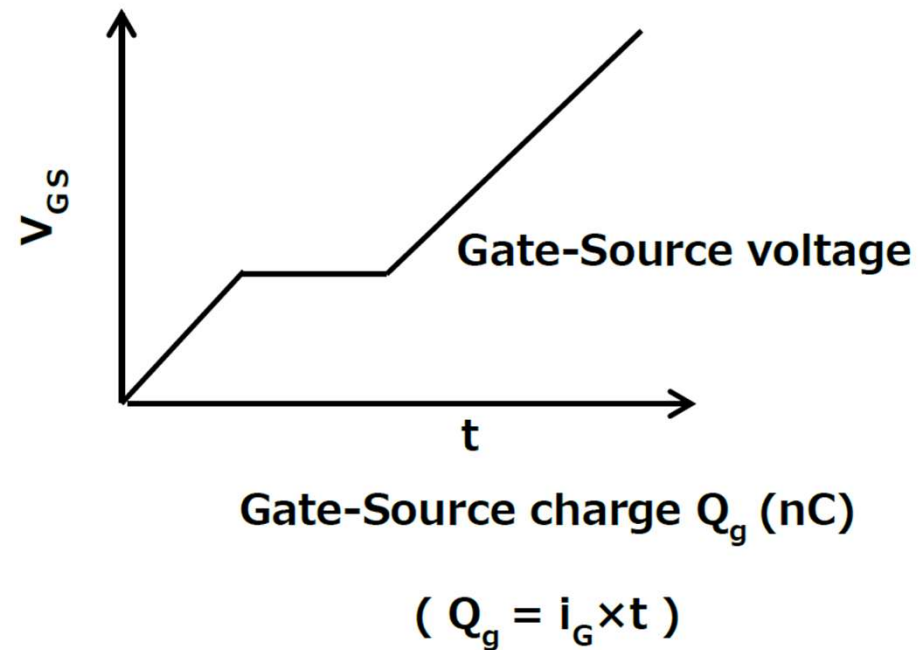
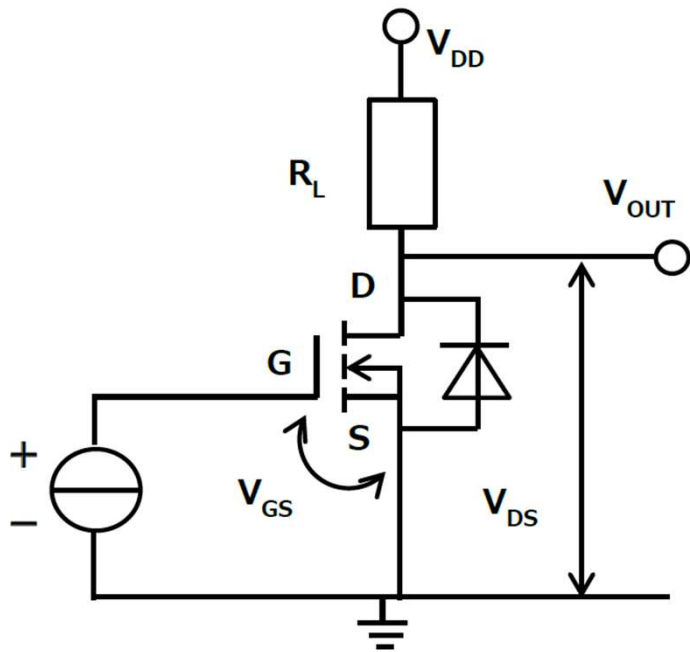
When V_{GS} exceeds V_{th} , the drain voltage starts to decrease and the CGD capacitance starts to get charged!



The gate voltage of a MOSFET does not increase unless its gate input capacitance is charged, and the MOSFET does not turn on until its gate voltage reaches the gate threshold voltage V_{th} . The gate threshold voltage V_{th} of a MOSFET is defined as the minimum gate bias required for creating a conduction channel between its source and drain regions.

GATE DRIVE

During the turn-on of a MOSFET, a current flows to its gate, charging the gate-source and gate-drain capacitances. Since the gate current is constant, the time axis can be expressed in terms of gate charge Q_g by multiplying time by constant gate current I_G . (The gate charge is calculated as $Q_g = I_G \times t$.)

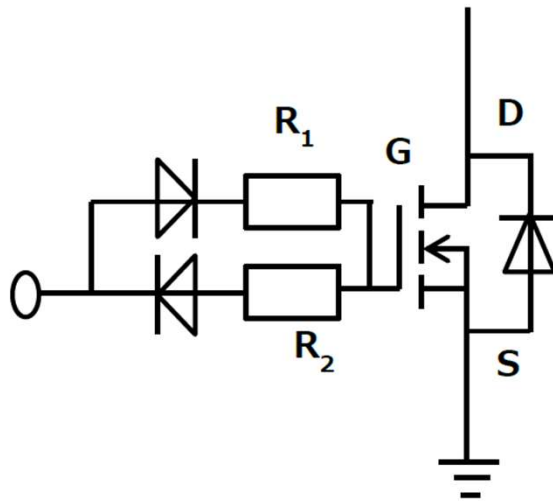


GATE DRIVE

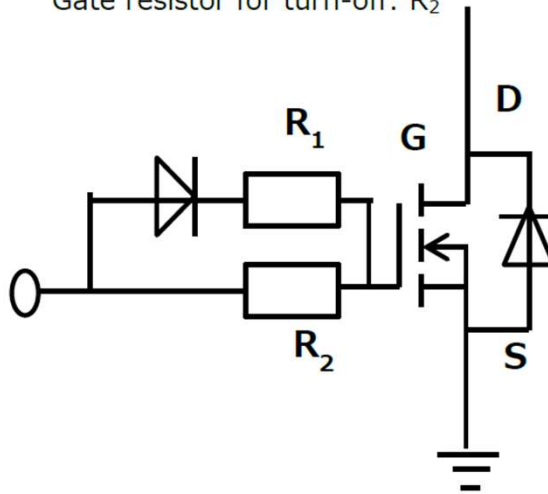
The switching speed varies with the gate resistor value. Increasing the gate resistor value slows the switching speed of a MOSFET and increases its switching loss. Reducing the gate resistor value increases the switching speed of a MOSFET but might cause a surge voltage to be applied between its drain and the source terminals due to the effects of wire stray inductance and other factors.

It is therefore necessary to select the optimal gate resistor. Sometimes, different gate resistors are used for the turn-on and turn-off of a MOSFET

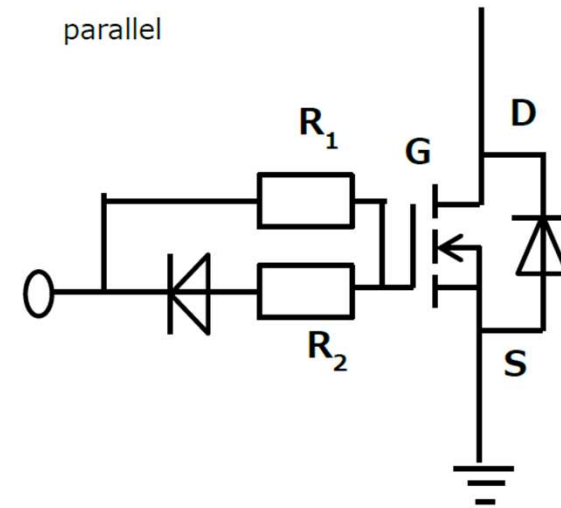
Gate resistor for turn-on: R_1
Gate resistor for turn-off: R_2



Gate resistors for turn-on: R_1
and R_2 in parallel
Gate resistor for turn-off: R_2



Gate resistor for turn-on: R_1
Gate resistors for turn-off: R_1 and R_2 in parallel

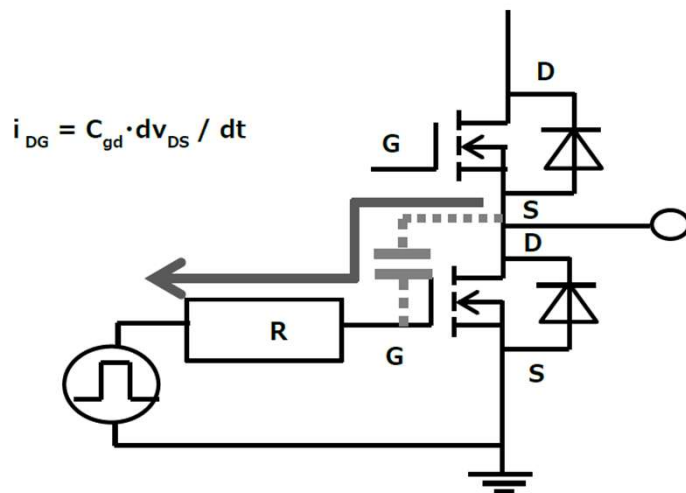


MILLER EFFECT and CLAMPING

One of the problems with MOSFETs is the parasitic turn-on (self-turn-on) phenomenon caused by their drain-gate capacitance. Upon **turn-off**, a sharp dv/dt develops between the source and the drain of a MOSFET. The resulting current flows to the gate via the drain-gate capacitance. As a result, a voltage drop that occurs across the gate resistor lifts the gate voltage. This current is calculated as:

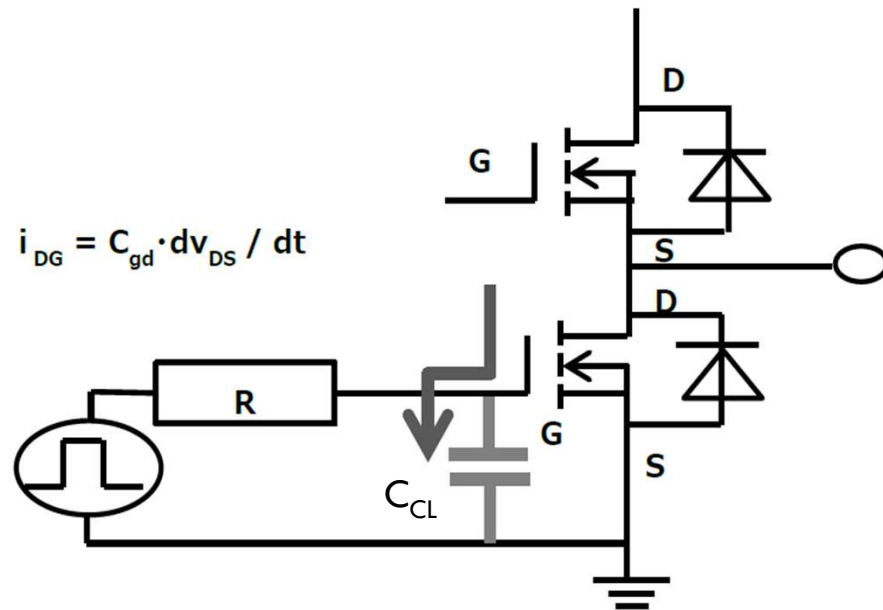
$$i_{DG} = C_{gd} \cdot dv_{DS} / dt$$

If dv/dt has a very sharp slope, a voltage is applied to the gate of a MOSFET, depending on the ratio of the gate-source capacitance to the gate-drain capacitance. If this occurs, self-turn-on might occur.



The turn-on gate resistor is usually made larger to limit the gate drive current and control di/dt (current change through the MOSFET) and EMI during switch-on. The turn-off gate resistor is kept smaller to discharge the gate faster, reducing switching losses by minimizing the overlap between drain current and drain-source voltage (and shootthrough).

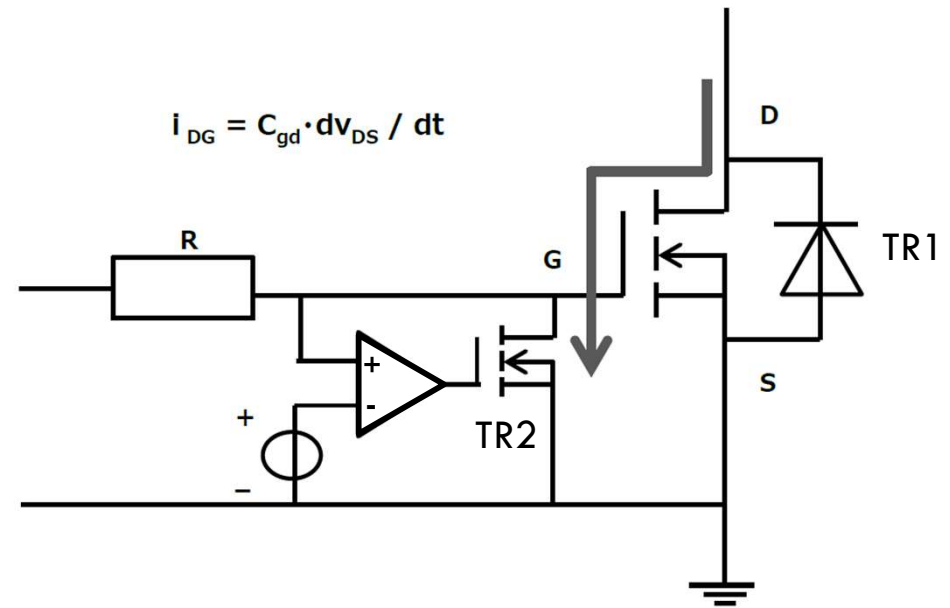
MILLER EFFECT and CLAMPING



Adding a capacitor across the gate and source terminals

Passive Clamping

C_{CL} directs the current i_{DG} to the ground, preventing a voltage rise at the gate of the LS transistor $\rightarrow$ forms a RC low pass filter at the gate, reducing the switching speed of the LS transistor! Therefore, C_{CL} shall be selected carefully!



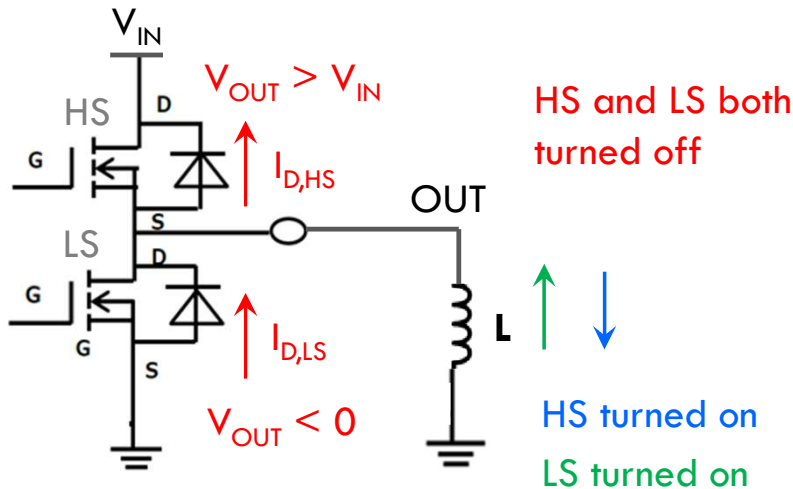
Miller clamp circuit

Active Clamping

If the voltage at the gate of TR1 rises, TR2 starts to conduct, pulling down the voltage at the gate of TR2 to 0V!

DEAD TIME – REMINDER (see the slides from previous weeks)

Shoot-Through Problem can be solved by introducing a so-called **Dead Time** in the PWM control signal



The output node «OUT» becomes floating during the dead time. What happens to the excessive energy of the inductor during dead time? → Current flow through the body diodes of the MOSFETs or freewheel diodes to V_{IN} or GND!

GATE RESISTOR

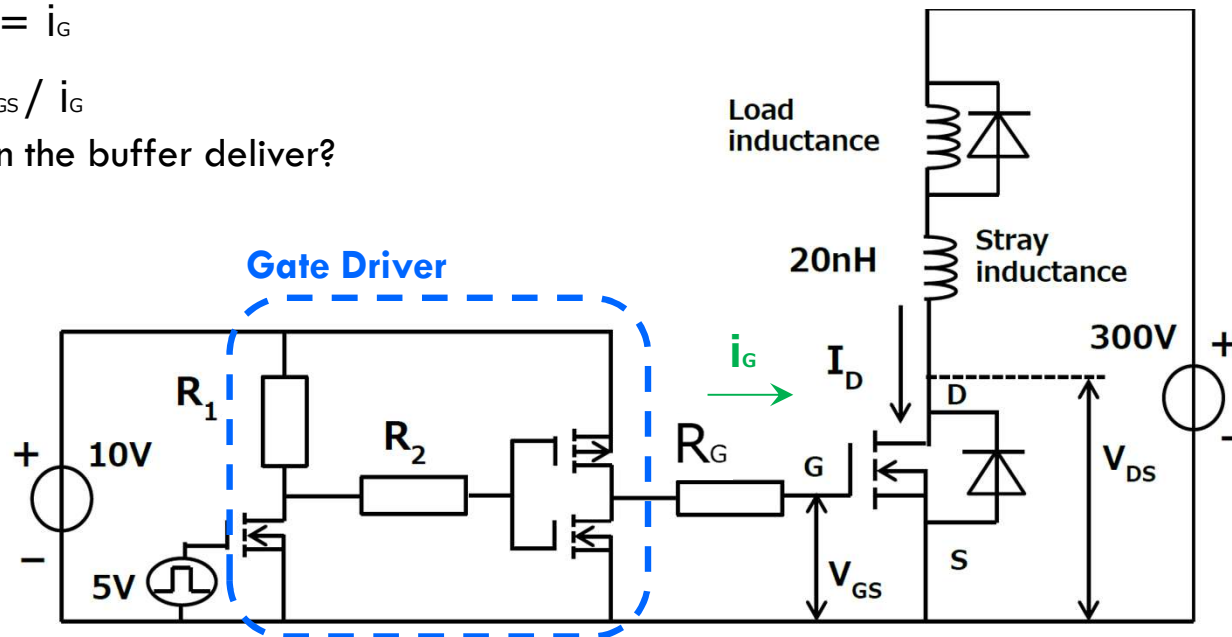
Generally, a resistor is connected to the gate terminal of a MOSFET. The purposes of the gate resistor include suppression of inrush current and a reduction in output ringing. A large gate resistor decreases the switching speed of a MOSFET. This results in an increase in power loss, a reduction in performance and potential heat issues. Conversely, a small gate resistor increases the switching speed of a MOSFET, which makes it susceptible to voltage surge and oscillation and therefore to device failure and damage. It is therefore important to optimize the MOSFET switching speed by adjusting the gate resistor value.

The gate rise time t_g and the gate resistor value R_G have the following relationships:

$$Q_g / t_g = i_g$$

$$R_G = V_{GS} / i_g$$

How much current can the buffer deliver?



Example: Low Side Transistor only, driving an inductive load

Example

A half-bridge converter uses an N-channel MOSFET whose total gate charge is $Q_g = 60 \text{ nC}$. The gate driver output voltage is 10 V. During turn-on, the designer wants the MOSFET to switch on in approximately $t_{on} = 120 \text{ ns}$. During turn-off, the target switching time is $t_{off} = 40 \text{ ns}$.

Assume: Linear charging and discharging of the gate. The gate current is approximately constant during switching

a) Calculate the required turn-on gate resistor R_{on} .

First, calculate the required gate current during turn-on:

$$I_{g,on} = \frac{Q_g}{t_{on}} = \frac{60 \text{ nC}}{120 \text{ ns}} = 0.5 \text{ A}$$

Now calculate the turn-on gate resistor:

$$R_{on} = \frac{V_g}{I_{g,on}} = \frac{10}{0.5} = 20 \text{ } \Omega$$

b) Calculate the required turn-off gate resistor R_{off} .

Gate current required during turn-off:

$$I_{g,off} = \frac{Q_g}{t_{off}} = \frac{60 \text{ nC}}{40 \text{ ns}} = 1.5 \text{ A}$$

Now calculate the turn-off gate resistor:

$$R_{off} = \frac{V_g}{I_{g,off}} = \frac{10}{1.5} \approx 6.8 \text{ } \Omega$$

Example

Same half-bridge with the following additional parameters: $Q_{gd} = 20 \text{ nC}$ and DC Bus Voltage $V_{DS} = 400 \text{ V} \rightarrow$ previous assumption of «*Linear charging and discharging of the gate. The gate current is approximately constant during switching*» is no more valid!

The designer allows a turn-on drain-source voltage fall time of $t_{dv/dt} = 100 \text{ ns}$.

Assume:

- * During the Miller plateau, almost all gate current flows into C_{gd}
- * Gate current is approximately constant during the Miller interval

a) Calculate the gate current during the Miller interval.

The Miller charge must be delivered within the allowed $t_{dv/dt}$

$$I_{g,Miller} = \frac{Q_{gd}}{t_{dv/dt}} = \frac{20 \text{ nC}}{100 \text{ ns}} = 0.2 \text{ A} = 200 \text{ mA}$$

b) Turn-on gate resistor

$$R_{on} = \frac{V_g}{I_{g,Miller}} = \frac{10\text{V}}{0.2\text{A}} = 50 \Omega$$

This resistor limits the gate current and therefore controls the Miller-region switching speed.

Example

In the first calculation ($20\ \Omega$), the resistor was sized using the total gate charge Q_g and a desired overall turn-on time (Assumption: linear charging). That approach answers the question: “How fast do I want the device to reach full enhancement?” It mainly controls timing and average gate current, but it does not directly control how fast the drain-source voltage changes. In the second calculation ($50\ \Omega$), the resistor was sized specifically for the Miller plateau, using Q_{gd} and an allowed dv/dt . This approach answers a different question: “How fast am I allowed to move V_{DS} during switching?” During the Miller plateau, almost all gate current flows into C_{gd} , and this current directly sets dv/dt . To limit dv/dt , the gate current must be lower, which requires a larger gate resistor.

The $20\ \Omega$ value controls overall turn-on speed based on Q_g . The $50\ \Omega$ value controls switching aggressiveness (dv/dt , EMI, stress) based on Q_{gd} .

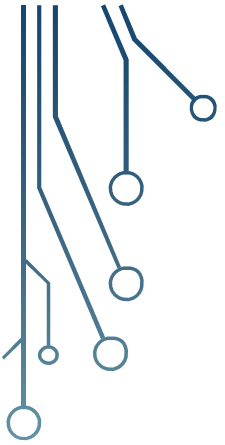
The drain-source voltage V_{DS} changes from 400 V to nearly 0 V during the Miller interval:

$$\frac{dv}{dt} = \frac{\Delta V_{DS}}{t_{dv/dt}} = \frac{400\text{ V}}{100\text{ ns}} = 4 \frac{\text{kV}}{\mu\text{s}}$$

This value is directly controlled by the gate current during the Miller plateau.

Example

During turn-off, the drain-source voltage rises rapidly. Due to the Miller capacitance C_{gd} , a displacement current flows from drain to gate. If the turn-off gate resistance is too large, this Miller current can raise the gate voltage and cause unintended re-turn-on of the MOSFET. A smaller turn-off gate resistance provides a low-impedance path to quickly remove this Miller-induced charge (in other words, low impedance to the ground via the small gate resistor and the nmos in the internal gate buffer), keeping the gate voltage below threshold and preventing shoot-through and excess switching loss. → please see the previous slides: Miller Effect and Clamping



References

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 - Analysis and Design of Analog Integrated Circuits”, Gray Mayer et al, 5th Edition, Wiley, 2009
 - Power Topologies Handbook, Markus Zehendner, Matthias Ulmann, Texas Instruments, 2020
 - Design and Application Guide of Bootstrap Circuit for High-Voltage Gate-Drive IC, AND9674/D, On Semiconductor, Application Note, 2021
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